

Question number	Answer	Additional guidance	Mark
7(a)(i)	An answer that includes the following points: • difference between systolic and diastolic pressure is 5.3 (kPa) (1) • person is healthy (because pulse pressure is greater than 3.75 kPa) (1)	ALLOW 35 / 35.3 ALLOW (because {35 /35.3} % is higher than 25%) CE applied to second point and comparison adjusted accordingly	(2)

Question number	Answer	Additional guidance	Mark
7(a)(ii)	An answer that includes the following points: • suitable estimated values chosen (1) • answer calculated (1)	ACCEPT values in range of (diastolic) 9.5 to 9.7 and (systolic) 14.8 to 15.0 to OR (systolic) 15 and (Diastolic) 10 11 / 11.2 / 11.3 / 11.4 / 11.5 OR 11.7 Correct answer with no working gains 2 marks	(2)

Question number	Answer	Additional guidance	Mark
7(a)(iii)	An explanation that includes the following points: - insufficient {oxygen / glucose} delivered to the {cells / tissues} (1) - credit an appropriate consequence (1)	ACCEPT oxygenated blood e.g. breathless, lack of energy, stroke, hypoxia, decrease in respiration, dizziness IGNORE death NB eg lack of oxygenated blood for cell respiration = 2 marks	(2)

Question number	Answer	Mark
7(b)(i)	D $\frac{8 \eta \lambda}{\pi r^4}$ The only correct answer is D. <i>A is incorrect because ΔP cancels out</i> <i>B is incorrect because ΔP cancels out</i> <i>C is incorrect because the equation is upside down</i>	(1)

Question number	Answer	Mark
7(b)(ii)	C radius of the blood vessel lumen The only correct answer is C <i>A is incorrect because ΔP cancels out</i> <i>B is incorrect because length is only to the power 1 and radius is to the power 4 and vessels can change their diameter</i> <i>D is incorrect because the thickness of the wall is not part of the calculation.</i>	(1)

Question number	Answer	Additional guidance	Mark
7(b)(iii)	An explanation that includes the following points: (inside of arteries lined with layer of unfolded) {flattened / smooth} {endothelial cells / endothelium} (1) - to reduce {friction / surface in contact with blood} (1)		(2)

Question number	Answer	Additional guidance	Mark
7(c)(i)	An explanation that includes the following points: • because elastic fibres (in wall of arteries) can {stretch / expand} (1) • therefore {widening the lumen / increasing the diameter (of the artery)} (1) • wall contains collagen to increase the strength (1)	IGNORE recoil DO NOT ACCEPT recoil	(2)

Question number	Answer	Additional guidance	Mark
7(c)(ii)	An explanation that includes the following points: • (if compliance is reduced) damage to the endothelium lining (1) • therefore, {cholesterol / (cholesterol) plaque} can build up (1) • therefore {narrowing / blocking} the blood vessel (1) • reducing {flow of blood / oxygen} to the heart {cells / tissues / muscle} (1)	N.B A reference to an artery must be made for 3 marks to be awarded ACCEPT atheroma IGNORE by clots	(3)